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# Combined effects of noise and air temperature on human neurophysiological responses in a simulated indoor environment

Ali Mohammad Abbasi [a], Majid Motamedzade [b], Mohsen Aliabadi [c,*], Rostam Golmohammadi [c], Leili Tapak [d]

[a] Department of Occupational Hygiene, School of Public Health, Ilam University of Medical Sciences, and Department of Occupational Health Engineering, Hamadan University of Medical Sciences, Iran
[b] Department of Ergonomics, School of Health and Research Center for Health Sciences, Hamadan University of Medical Sciences, Hamadan, Iran
[c] Center of Excellence for Occupational Health, Occupational Health & Safety Research Center, Hamadan University of Medical Sciences, Hamadan, Iran
[d] Department of Biostatistics, School of Public Health and Modeling of Non- Communicable Diseases Research Center, Hamadan University of Medical Sciences, Hamadan, Iran

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ABSTRACT

The aim of the present study was to evaluate the combined effects of noise and air temperature on the human body neurophysiological responses. This study was conducted on 35 male students, who were exposed to four different air temperatures (18 °C, 22 °C, 26 °C, and 30 °C) and two noise levels (55 dBA and 75 dBA) in eight sessions in a simulated indoor environment. The mean values of accuracy and time of response to stimuli in N-back test as well as neurophysiological responses were measured. In the studied experiment configurations, with increasing air temperature and noise, the working memory and neurophysiological responses were disturbed. The results indicated the significant effect of noise on working memory, as compared with that of air temperature. The effects of air temperature on heart rate, respiratory rate as well as theta and alpha bands were more significant than the impact of noise. The combined effects of noise and air temperature were more significant than the influence of each of them alone. In the presence of high noise levels, the increase in air temperature did not worsen the response accuracy. However, in the presence of high noise level, the rise in air temperature aggravated the mean value of neurophysiological responses. Overall, noise has a greater effect on working memory, while the air temperature can disturb neurophysiological responses in a more profound way.

## 1. Introduction

People frequently spend most of their time in closed spaces such as home as well as commercial or industrial places. Therefore, the quality of indoor environments can affect their comfort and performance (Tiller et al., 2010). Some studies have shown the significant effects of the environmental factors on employees' cognitive performance and job satisfaction (Golmohammadi et al., 2017; Clausen et al., 2013). Noise and air temperature are believed to be the most important indoor environmental factors that affect human's health and work performance in mental activities (Liebl et al., 2012; Lan et al., 2011).

Indoor environmental factors can individually affect the health and comfort of employees, while noise and air temperature can also have combined effects on these individuals (Hemmatjo et al., 2017; Lan et al., 2011a,b; Varjo et al., 2015). In recent years, some researchers have designed experimental studies to investigate the individual effects of noise and air temperature on cognitive performance using advanced test methods (Kaczmarska and Luczak, 2007). As reported by Li et al. (2014), when subjects were exposed to noise for 5 min, the sum of the relative average power of electroencephalogram (APEEG) of theta wave and the relative APEEG of alpha wave increased. The subjects' annoyance level based on the sum of relative APEEG of h wave was also aggravated under this noise exposure scenario. Lan et al. (2009) showed that air temperature had diverse effects on staff cognitive performance depending on the type of the tasks they were engaged with. Gaoua et al. (2012) reported that the increase in body temperature is apparently a sufficient physiological response that alters the subjective state of participants and disturbs decision-making. Lan et al. (2010) also focused on body physiological responses to hot and humid environments, while ignoring the influential factors. Cui et al. (2013) found that the optimum temperature

* Corresponding author..
*E-mail addresses:* mohsenaliabadi2@gmail.com, mohsen_ohse@yahoo.com (M. Aliabadi).

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range for mental performance was from 22 to 26 °C. They also reported that the discomfort caused by warm environment can disturb both motivation and performance.

Note that, in a real workroom, people are exposed to all these factors simultaneously. Further, the environmental parameters are all interrelated, and human comfort is a complex state involving an occupant's sensations of all these factors (Zhang et al., 2010; Huan et al., 2004; Pellerin and Candas, 2003). ASHRAE guideline10 (2016) concentrated on interactions among the indoor environmental factors and the possible ways through which different factors might affect each other. In this regard, some studies have tried to investigate the interaction between noise and air temperature on human body responses. Piasecki et al. (2017) showed that people's perceptions of thermal conditions are not altered through short-term exposure to different noise levels. However, thermal conditions affect the subjects' noise perceptions depending on the types of noise and relative air humidity. Pellerin and Candas (2004) also reported that noise may alter human's thermal comfort in warm environments. They observed that a temperature change 1 °C had the same effect as 2.6 dB variation in noise. Yang et al. (2018) reported that short-term exposure to moderate noise cannot affect thermal sensations; nonetheless, air temperature and humidity levels impact the perceptions of loudness and annoyance levels as well as acoustic preferences. Varjo et al. (2015) investigated how irrelevant speech and air temperature together affect the mental performance and environmental comfort among office staffs. While being exposed to a great deal of irrelevant speech and high air temperature, they observed that the employees' cognitive performance diminished especially in mental tasks. Yang and Moon (2018) also noted that the effects of noise on thermal comfort and impacts of thermal sensation based on predicted mean vote (PMV) on acoustic comfort and noise annoyance were statistically significant.

Torresin et al. (2018) stated that humans are exposed simultaneously to acoustic, thermal as well as visual and air quality stimuli in indoor environments. They confirmed that research is still ongoing to clarify the nature of the interaction and derive practical indications for designers and practitioners. Huang et al. (2012) stated that the satisfaction levels of both air temperature and noise have one-vote veto power over the satisfaction level of the other indoor environment factors. Choi et al. (2015) reported that it is necessary to investigate the mechanism behind indoor environmental factors influencing neurophysiological responses of the human body. Chao et al. (2013) emphasized that the combined effects of noise and air temperature are still unclear; therefore, the impact of various exposure times, levels and types of noise as well as dynamic thermal conditions warrant further studies.

Less information is available to draw reliable conclusions on the effects of simultaneous exposure to noise and relatively extreme air temperature on cognitive performance and mental health. Until now, no consistent results have been found about these topics (Yang and Moon, 2018). The current experimental study aims to explore the plausible mechanism linking air temperature and noise to human comfort and performance through analyzing a wide range of advanced neurophysiological response parameters via electroencephalography and electrocardiography in a simulated office workroom. The current study is an attempt to understand the interaction in detail. In this way, the effect size data about different exposure scenarios can help experts analyze the priorities in the design of indoor environments and find constructive solutions to control environmental variables.

## 2. Materials and methods

### 2.1. Participants

Thirty-five healthy male students from Hamadan University of Medical Sciences were selected as the participants. Their age, height, weight, and body mass index (BMI) were 22.9 ± 2.9 year, 178.0 ± 4.9 cm, 72.9 ± 10.3 kg, and 23.1 ± 3.2 respectively. The participants were screened for visual and hearing health. The visual test was carried out using E chart, while the hearing health was tested using an audiometer (MEVOX ASB15). The participants with a minimum average hearing threshold of 25 dBHL were enrolled in the study. The subjects with a history of neuropsychiatric diseases and/or drug use that affected their nervous system were excluded from the study. In addition, the subjects were examined for right-handedness and color blindness. They were asked to get enough sleep before the test and were recommended not to smoke and not to use caffeine or any other stimulants 12 h before the experiments. All participants were paid in order to enhance their motivation and to encourage them to perform the experiments seriously and correctly. It seems that, due to the nature of the study experiments, including repeated measurements, the same salary for participation could prevent missing participants during different sessions. As a result, all of them successfully completed the experimental sessions. It was also observed that the participants attended the sessions on time and performed the tests continuously. Further, in the interview before each experiment, they stated their verbal satisfaction with tests at the different sessions. This study was approved by the Ethics Committee of Hamadan University of Medical Sciences (ethic code: IR.UMSHA.REC.1395.457). After confirmation, informed consent was obtained from the participants.

### 2.2. Experimental set up

The study was conducted in an air conditioning chamber with the dimensions of L × W × H = 3.70 × 2.40 × 2.70 m. The walls of the chamber were covered with bright colored polyurethane foam absorbent to make it psychologically appropriate to resemble indoor wall surfaces of typical offices (Dzulkifli and Mustafar, 2013). In this room, there was a table, a chair, and a personal computer to resemble an office's workstation as shown in Fig. 1. Further, the minimum level of room lighting was 300 lux (Huang et al., 2012), supplied by two LED lights. The indoor of the chamber was carefully furnished and finished to resemble a typical normal office environment so that the participants feel same about the simulated indoor environment.

It was possible to set and fix the chamber's thermal conditions using an air-conditioner system. The relative humidity was kept constant at 40% and air velocity was set at 0.15 m/s during all experiments. The participants were exposed to different air temperatures including 18 °C (cool), 22 °C (relatively neutral temperature), 26 °C (a little higher than the natural temperature), 30 °C (which is not too high for a naturally ventilated office environment under summer conditions), along with low frequency noise levels including 55 dBA (typical background noise level) and 75 dBA (resembling noisy office workplace). The thermometer sensors installed on the room's wall continuously monitored the



**Fig. 1.** Experimental setup in the simulated indoor environment.

thermal conditions including air temperature, and relative humidity during the experiment.

The frequency spectrum of the simulated noise in one octave band is presented in Fig. 2. Note that low-frequency noise is one of the most harmful and annoying factors that frequently appear in human working and living environments (Kaczmarska and Luczak, 2007). Fig. 2 presents the frequency spectrum of a recorded fan noise similar to daily indoor noise sources, which was played by two 10 W speakers and one 8 W subwoofer (Mingo mg 202). The speakers were placed on both sides of a monitor 0.5-m apart from each other. Thus, under the designed conditions, all the participants were exposed to similar noise levels. The noise levels perceived by listeners' ears as well as the air temperature around their body were monitored every 10 min to check the designed exposure scenario condition.

The participants were asked to select a specific day and time to perform the experiments. They were also requested to wear a clothing ensemble with an insulation value of 0.72 clo. Their clothes included a long-sleeved shirt, thin cotton trousers, calf-length socks, and shoes, which are representative of a typical office employee dress. The participants were exposed to eight environmental conditions on eight separate days with each experimental session lasting 50 min. Note that the participants were individually tested in each session. The experiment was carried out using a within-participants design, where all participants were tested under eight experimental conditions, thus acting as their own controls.

Before the participants entered the chamber, the neurophysiological responses (baseline) were measured in a room considered near the chamber for 10 min. Next, the participants entered the chamber reading books or playing games for 5 min to adapt to the indoor environment. Following the adaptation, they spent about 5 min on performing the N-back tests (low workload), during which their physiological responses were measured. After completing the tests, 5 min were allocated to participants' rest. The participants spent again about 5 min on performing the N-back tests (medium workload) and simultaneously their physiological responses were measured. After completing the second N-back tests, they were again given 5 min to have a rest. Finally, the participants spent about 5 min on performing the third round of N-back tests (high workload), while their physiological responses were measured. Ten minutes later, their neurophysiological responses were again measured. The flowchart of different steps of the experiments is presented in Fig. 3.

In the current study based on a within-participants design, the possible carryover effects were partially controlled by the random exposure (instead of sequential exposure) to different scenarios (combined air temperature and noise level) and also to different N-back tests. Further, it is argued that if the subjects are fully trained and familiar enough with the tasks before the experiment, the learning effect can be approximately removed (Lan et al., 2009). Thus, a week before starting the tests, the participants practiced the N-back to get familiar with the test procedure as well as the N-back software. In this regard, the subjects were given two separate 30-min training sessions, followed by the examination of their performance. If a defect was observed, the participant was again given the opportunity to practice until the defect was resolved. In the second training session, a pilot study was performed in order to compare different stimulated workloads (N-back) with the perceived workloads for 10 participants using NASA TLX and EEG. These trail experiments showed no significant differences between stimulated workloads and the perceived workloads. In addition, following the pilot study, exposure orders to different workloads were checked in six scenarios based on EEG. There were not significant differences between the different exposure orders for the 10 participants.

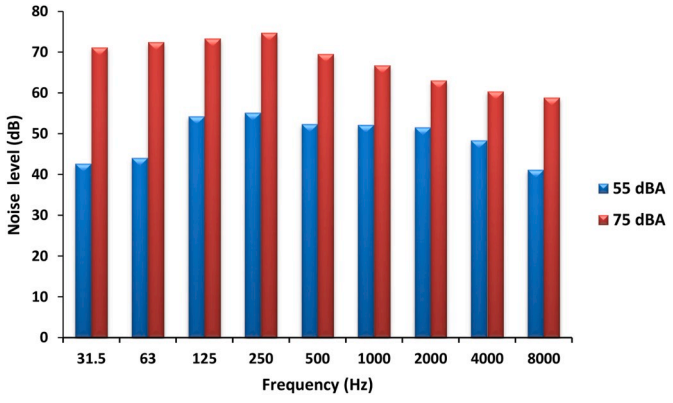



**Fig. 2.** Frequency spectrum of noise sources in one octave band.

All participants sat on the chair during the tests whereby the workload was increased step by step, from low to high. Each task lasted for 5 min and there was 2500 ms intervals between each stimulus. The participants were presented a total of 120 stimuli, which were implemented at all levels of the task simultaneously under the same air temperature and noise level. In order to prevent errors in the measurements, the participants were asked to have minimum movement and focus on the given task. After installing ECG electrodes, three signal levels were measured before the exposure and during the exposure to each of the air temperature and noise levels through N-back test while performing three workloads (low, medium, and high) for 5 min. Also, before and immediately after the exposure to any air temperatures and noise levels, the electrical activities of the brain and eyes were measured (Abbasi et al., 2018).

### 2.3. Types of measurements

#### 2.3.1. Working memory

N-back test was employed at three cognitive levels (low workload (n = 1), medium workload (n = 2), and high workload (n = 3)) to evaluate the working memory. N-back task is a cognitive performance measurement task with an executive function originally introduced by Kirchner (1958). Since this task involves saving and manipulating cognitive information, it is known as a suitable method to measure the working memory performance. In this test, a number of visual stimuli (from 1 to 9) randomly appear on the computer screen and the subjects must respond to each of them in three different conditions with various loads of working memory. Under the low workload condition, the participants compared each visual stimulus with the previous one; if the two consecutive images were the same, they should have pressed the question mark button (?) on the computer keyboard; else the Z Button. The participants performed the same task in the end-test. Under the medium workload condition, the subjects should have compared the stimulus with the two previous stimuli; if the stimuli were similar, they were supposed to press question mark button (?); else the Z Button. Finally, under the high workload condition, the subjects were asked to compare each stimulus with the three previous stimuli; if the stimuli were similar, they were supposed to press the question mark (?); else the Z Button (Kirchner, 1958; Savage and Goghari, 2016).

The results of the N-back test show the number of correct and incorrect answers as well as the average response time (Jaeggi et al., 2010). In this study, the stimuli were random numbers from 1 to 9. The number of correct answers (accuracy) and the average response time (in milliseconds) were calculated by N-back software under the four different conditions of exposure under different air temperatures (18 °C, 22 °C, 26 °C, and 30 °C) and noise levels (55 dBA and 75 dBA).

#### 2.3.2. Neurophysiological responses

Nexus-4 device (manufactured by Mind Media Netherlands), which is equipped with Bluetooth technology was used to measure the neurophysiological responses. It is suitable for collecting signals of EEG, electrocardiograph (ECG), electromyograph (EMG), and electrooculograph (EOG). It is also able to monitor and measure the skin temperature (TMP), respiratory rate (RSP), galvanic skin response (GSR), and blood

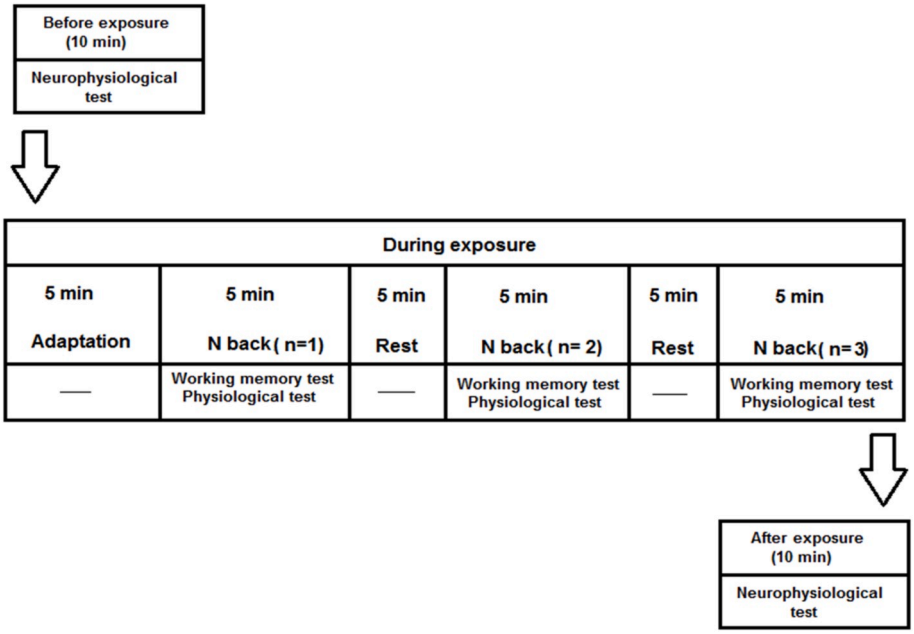



**Fig. 3.** Flowchart of experiment setup.

volume pulse (BVP). In order to store and monitor data online, the collected signals are wirelessly transferred into a computer using Bluetooth technology. Data processing, digital filtering, process reporting, and preliminary statistical analysis were performed by software (BioTrace software®, Mind Media BV, Roermond -Herten, The Netherlands) (Abbasi et al., 2018). The data on the operational channel was used to measure ECG signals at 1024 Hz. ECG signals are recorded by installing three Ag-Agcl electrodes on both sides of the sternum bone and left side armpit on the sixth rib. The use of the ECG signal on the Nexus is based on the heart rate (HR) and heart rate variability (HRV) measurements. Heart rate variability was evaluated based on ECG values measured under different working conditions (low, medium, and high), different air temperatures (18 °C, 22 °C, 26 °C, and 30 °C), and noise levels (55 dBA and 75 dBA). HRV analysis was carried out at low frequency (LF) (0.04–0.15 Hz) and high frequency (0.15–0.4 Hz) ranges (Sayers, 1973). The low to high frequency ratio (LF/HF) usually indicates the activity of the sympathetic nervous system (Rosenberg et al., 2017). After recording ECG signals, the average HR and LF/HF were measured for 5 min under different workloads, air temperatures, and noise levels before and during exposures. Further, Nexus-4 sensors fastened around the body like a belt, were used to measure the relative expansion of the abdomen or chest during inhalation and exhalation. A large size sensor was used around the abdomen, and it was naturally positioned around the navel. To achieve the best result, a layer of the shirt was placed between the skin and the sensor. There was also a suggested sample rate for the operating channel to record RSP on Nexus 4 using the Sensor (NX-RSP1A) 32 SPS. A frequency of 32 Hz was used in the C channel. The belt was fastened around the abdomen and RSP signals were recorded for 5 min under different workloads (low, medium, and high), air temperatures (18 °C, 22 °C, 26 °C, and 30 °C), and noise levels (55 dBA and 75 dBA) at two steps (before and during the test). The operating channel was used to measure 256 Hz-EEG and EOG indices. Nexus is one of the few systems that utilizes active noise cancellation (movement artifacts and external interactions) for physiological signals. To measure electrophysiological indices, Nexus uses carbon-coated cable with active shielding, which provides clear signals without any movement artifacts. It also employs active noise-cancelling technology, which reduces the movement artifacts and external interactions, and also provides signals with a good quality. The channel A was used in Nexus to record EEG based on the international10-20 the place of electrodes. To record EEG signals, reference electrodes were placed on the mastoid bone of the left ear (black) while the ground electrode was placed on the mastoid bone of the right ear (white) with red electrode being placed on the Cz site. In this study, we used NuPrep gel (for skin preparation) and EEG 10–20 (Ten20) paste to fix the electrodes. Furthermore, the second channel (input B) was used to check EOG (Blink Measurement and Eye Movement Measurement). To record EOG (theta band), two small ECG electrodes were used: one placed above and the other below the left eye (Vertical EOG). Using the data recorded by EEG and EOG signals, the mean theta amplitude (4–8 Hz), alpha amplitude (8–12 Hz), beta amplitude (12–15 Hz), and average range of eye electrical activity were measured under different air temperatures and at noise levels at two steps (before and immediately after the test) (Abbasi et al., 2018).

#### 2.3.3. Statistical analysis

The data were analyzed using repeated measure analysis of variance. To evaluate the normal distribution of the data at different levels, the Mauchly's Test of Sphericity was used. When the sphericity hypothesis was rejected, the Huynh-Feldt's analysis was used to correct the degree of freedom. Also, the Tukey's Test for Post-Hoc analysis was carried out to examine the difference between the means. The effect size (ES) was also reported. SPSS-Ver24 was employed for statistical analysises. The significant level for all tests was set at 5%.

## 3. Results

### 3.1. Working memory

Table 1 presents the accuracy and speed of response to stimuli in working memory experiments at different workloads under different noise levels and air temperatures. The results revealed a significant difference between accuracy in low, medium, and high workloads under different air temperatures at noise level of 55 dBA ($p < 0.05$). A significant difference is observed between the mean values of accuracy only under low workload and during exposure to a noise level of 75 dBA under different air temperatures ($p < 0.05$). Table 2 also shows multiple comparisons of response accuracy and response speed in three workloads under different noise levels and air temperatures. Some significant and non-significant differences are observed for the accuracy and speed of response in the multiple comparison of air temperatures. However, it

**Table 1**
Means ± SD of response accuracy and response speed (average response time) of working memory test in exposure to different noise levels and air temperatures.

| Noise levels | Parameters | Air temperatures | | | | | |
|---|---|---|---|---|---|---|---|
| | | 18 °C | 22 °C | 26 °C | 30 °C | ES | p-value |
| **55 dBA** | Accuracy n = 1 | 87.05 ± 9.93 | 89.24 ± 12.22 | 84.22 ± 12.22 | 78.11 ± 13.86 | 0.151 | 0.001 |
| | Accuracy n = 2 | 77.80 ± 11.57 | 85.69 ± 6.36 | 80.20 ± 11.00 | 74.02 ± 12.97 | 0.185 | 0.001 |
| | Accuracy n = 3 | 73.22 ± 12.73 | 82.52 ± 7.12 | 77.88 ± 11.80 | 71.99 ± 12.04 | 0.155 | 0.001 |
| | Response time n = 1 | 430.17 ± 160.29 | 449.49 ± 130.04 | 470.54 ± 114.04 | 459.18 ± 136.41 | 0.016 | 0.636 |
| | Response time n = 2 | 485.02 ± 150.06 | 530.97 ± 1530.79 | 525.20 ± 160.27 | 506.63 ± 135.26 | 0.026 | 0.437 |
| | Response time n = 3 | 523.29 ± 159.89 | 575.60 ± 156.35 | 569.71 ± 83.02 | 563.89 ± 136.60 | 0.036 | 0.297 |
| **75 dBA** | Accuracy n = 1 | 76.85 ± 10.62 | 78.22 ± 13.29 | 77.74 ± 13.56 | 68.28 ± 14.57 | 0.117 | 0.005 |
| | Accuracy n = 2 | 63.45 ± 13.97 | 67.05 ± 13.53 | 66.11 ± 14.78 | 65.05 ± 16.60 | 0.011 | 0.767 |
| | Accuracy n = 3 | 58.31 ± 12.79 | 60.88 ± 18.25 | 59.02 ± 12.73 | 57.31 ± 13.89 | 0.011 | 0.762 |
| | Response time n = 1 | 440.74 ± 150.44 | 430.80 ± 68.99 | 458.60 ± 91.69 | 545.51 ± 135.02 | 0.163 | 0.001 |
| | Response timen = 2 | 490.51 ± 122.71 | 448.80 ± 88.71 | 539.49 ± 92.17 | 516.77 ± 135.24 | 0.112 | 0.009 |
| | Response time n = 3 | 540.14 ± 140.68 | 489.94 ± 73.08 | 563.20 ± 89.04 | 569.29 ± 133.60 | 0.098 | 0.018 |

**Note.** Accuracy: Percentage of correct answers, Response time: Average response rate to the stimuli, n = 1, n = 2 and n = 3, respectively: low workload, medium workload, high workload, ES: Effect size.

**Table 2**
Multiple comparisons of response accuracy and response speed in exposure to different noise levels and air temperatures.

| Noise | Parameters | 18 °C | | | 22 °C | | | 26 °C | | | 30 °C | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 22 °C | 26 °C | 30 °C | 18 °C | 26 °C | 30 °C | 18 °C | 22 °C | 30 °C | 18 °C | 22 °C | 26 °C |
| **55 dBA** | Accuracy n = 1 | NS | NS | **0.008** | NS | NS | **0.001** | NS | NS | NS | **0.008** | **0.001** | NS |
| | Accuracy n = 2 | **0.001** | NS | NS | **0.001** | **0.011** | **0.001** | NS | **0.011** | **0.037** | NS | **0.001** | **0.037** |
| | Accuracy n = 3 | **0.001** | NS | NS | **0.001** | NS | **0.001** | NS | NS | NS | NS | **0.001** | NS |
| | Response time n = 1 | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| | Response time n = 2 | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| | Response time n = 3 | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| **75 dBA** | Accuracy n = 1 | NS | NS | **0.006** | NS | NS | **0.005** | NS | NS | **0.010** | **0.006** | **0.005** | NS |
| | Accuracy n = 2 | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| | Accuracy n = 3 | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| | Response time n = 1 | NS | NS | **0.004** | NS | NS | **0.001** | NS | NS | **0.011** | **0.004** | **0.001** | **0.011** |
| | Response time n = 2 | NS | **0.029** | NS | NS | **0.001** | **0.014** | **0.029** | **0.001** | NS | NS | **0.014** | NS |
| | Response time n = 3 | NS | NS | NS | NS | **0.001** | **0.004** | NS | **0.001** | NS | NS | **0.004** | NS |

**Note.** The reported values indicating significant difference (P < 0.05) and NS indicating no significant. n = 1, n = 2 and n = 3 indicating low, medium and high workloads, respectively.

can be seen that upon air temperature rise from 22 °C, the accuracy of the responses reduced. Further, the accuracy of responses under different air temperatures was higher at 55 dBA than at 75 dBA in different workloads. With elevation of air temperature from 22 °C, the speed of response increased only at 75 dBA. Table 3 and Table 4 show the significant effect of noise (75 dBA, 22 °C) along with different effect sizes of 0.351, 0.590 and 0.565 on the response accuracy, as compared with air temperature (55 dBA, 30 °C) along with effect sizes of 0.308, 0.390 and 0.379 for three workloads, respectively. It is observed that the combined effect of high noise and high air temperature condition (75 dBA at 30 °C) on response accuracy was stronger than the individual effects of high noise condition (75 dBA at 22 °C) and high air temperature condition (30 °C at 55dBA). In other words, the combined effects of noise and air temperature were more significant than the influence of each of them alone.

However, with increasing noise and air temperatures, the speed of response to stimuli did not show a uniform trend. Likewise, the speed of responses under different air temperatures did not have any considerable trend at 55 dBA, as compared with 75 dBA, in different workloads. Concerning the combined effect of noise and air temperature on the working memory, the effect size of increasing air temperatures on response accuracy at 55 dBA was generally higher than that 75 dBA. In other words, in the presence of a high noise level of 75 dBA, the effect of air temperature elevation on the accuracy did not worsen. It can be concluded that high noise neutralized the effect of increasing air temperatures on the response accuracy.

As shown in Tables 3 and 4, similar results were observed when evaluating the effect size of increasing noise levels on the response accuracy under two different air temperatures (22 °C and 30 °C). The results can be attributed to the significant interaction effect of exposure to

**Table 3**
Pair wise comparison of the combined effects of noise levels and air temperatures on the accuracy and speed of responses.

| Noise levels- air temperatures | Accuracy n = 1 | | | Accuracy n = 2 | | | Accuracy n = 3 | | |
|---|---|---|---|---|---|---|---|---|---|
| | MD | ES | p | MD | ES | p | MD | ES | p |
| 75dBA-22°Cvs.55dBA-22 °C | −11.01 ± 2.26 | 0.351 | 0.001 | −18.63 ± 2.66 | 0.590 | 0.001 | −21.64 ± 3.25 | 0.565 | 0.001 |
| 75dBA-30°Cvs.55dBA-30 °C | −9.82 ± 2.49 | 0.314 | 0.001 | −9.00 ± 2.43 | 0.287 | 0.001 | −14.65 ± 2.72 | 0.461 | 0.001 |
| **Noise levels- air temperatures** | **Response time n=1** | | | **Response time n=2** | | | **Response time n=3** | | |
| | **MD** | **ES** | **p** | **MD** | **ES** | **p** | **MD** | **ES** | **p** |
| 75dBA-22°Cvs.55dBA-22 °C | −18.68 ± 23.37 | 0.018 | 0.430 | −82.17 ± 29.72 | 0.184 | 0.009 | −85.67 ± 30.22 | 0.191 | 0.028 |
| 75dBA-30°Cvs.55dBA-30 °C | 85.68 ± 23.35 | 0.302 | 0.001 | 10.14 ± 3.80 | 0.173 | 0.012 | 5.40 ± 1.66 | 0.253 | 0.003 |

**Note.** MD: Mean of difference, SE: Standard error, ES: Effect size.

**Table 4**
Pair wise comparison of the combined effects of different air temperatures and noise levels on the accuracy and response speed.

| Accuracy n = 3 | | | Accuracy n = 2 | | | Accuracy n = 1 | | | Air temperatures- noise levels |
|---|---|---|---|---|---|---|---|---|---|
| p | ES | MD | p | ES | MD | p | ES | MD | |
| 0.669 | 0.005 | −125.00 ± 2.61 | 0.177 | 0.053 | −3.77 ± 2.73 | 0.008 | 0.188 | −8.94 ± 3.18 | 30°C-55dBAvs.18°C-55dBA |
| 0.001 | 0.379 | −10.60 ± 2.32 | 0.001 | 0.390 | −11.66 ± 2.50 | 0.001 | 0.308 | −11.12 ± 2.85 | 30°C-55dBAvs.22°C-55dBA |
| 0.077 | 0.089 | −5.91 ± 3.24 | 0.032 | 0.121 | −6.17 ± 2.85 | 0.052 | 0.109 | −6.11 ± 3.04 | 30°C-55dBAvs.26°C-55dBA |
| 0.749 | 0.003 | −1.00 ± 3.10 | 0.670 | 0.005 | 1.57 ± 3.65 | 0.006 | 0.199 | −8.57 ± 2.94 | 30°C-75dBAvs18°C-75dBA |
| 0.001 | 0.405 | −1.31 ± 0.27 | 0.027 | 0.139 | −2.94 ± 1.27 | 0.001 | 0.340 | −10.25 ± 2.45 | 30°C-75dBAvs22°C-75dBA |
| 0.601 | 0.008 | −1.71 ± 3.25 | 0.760 | 0.003 | −1.06 ± 3.51 | 0.010 | 0.179 | −9.45 ± 3.49 | 30°C-75dBAvs26°C-75dBA |
| **Response time n=3** | | | **Response time n=2** | | | **Response time n=1** | | | **Air temperatures- noise levels** |
| **p** | **ES** | **MD** | **p** | **ES** | **MD** | **p** | **ES** | **MD** | |
| 0.001 | 0.036 | 48.19 ± 23.30 | 0.541 | 0.011 | 21.60 ± 34.99 | 0.490 | 0.016 | 29.11 ± 38.91 | 30°C-55dBAvs.18°C-55dBA |
| 0.730 | 0.004 | −11.71 ± 33.60 | 0.490 | 0.014 | −24.34 ± 34.89 | 0.659 | 0.006 | −12.51 ± 28.10 | 30°C-55dBAvs.22°C-55dBA |
| 0.817 | 0.002 | −5.82 ± 25.05 | 0.524 | 0.012 | −18.57 ± 28.85 | 0.742 | 0.003 | −10.71 ± 32.33 | 30°C-55dBAvs.26°C-75dBA |
| 0.391 | 0.022 | 26.14 ± 33.52 | 0.412 | 0.020 | 26.25 ± 31.62 | 0.004 | 0.215 | 38.77 ± 34.31 | 30°C-75dBAvs.18°C-75dBA |
| 0.002 | 0.248 | −50.57 ± 15.10 | 0.146 | 0.061 | −24.48 ± 16.45 | 0.041 | 0.117 | −30.68 ± 14.46 | 30°C-75dBAvs.22°C-75dBA |
| 0.813 | 0.002 | 6.06 ± 25.57 | 0.444 | 0.017 | −22.71 ± 29.73 | 0.011 | 0.176 | 86.91 ± 32.20 | 30°C-75dBAvs.26°C-75dBA |

**Note.** MD: Mean of difference, SE: Standard error, ES: Effect size.

noise and air temperature on the working memory. Note that based on the effect size of three workload levels on working memory, work complexity does not interact with noise and air temperature. The results of the tests under the air temperature of 18 °C showed that all the aspects of working memory performance reduced, as compared with the tests performed under the air temperature of 22 °C. This can be due to some deviation from the optimum value of arousal in response to mild/moderate cold exposure, which slightly decreased the memory performance.

### 3.2. Neurophysiological responses

Table 5 presents participants' neurophysiological responses at various workloads under different noise levels and air temperatures. No significant difference is observed between the mean heart rates, LF/HF, and RSP under different workloads at different air temperatures with a noise level of 55 dBA ($P > 0.05$). It can be potentially related to the lack of arousal in low background noise level. On the other hand, the results showed significant differences between the mean heart rates, LF/HF ratio, and RSP under different workloads at various air temperatures with a noise level of 75 dBA ($p < 0.05$). This can be attributed to increase in the noise level stimulated arousal. Table 6 also presents multiple comparisons of the physiological responses in three workloads during exposure to different noise levels and air temperatures. There are some significant and non-significant differences of the physiological responses in the multiple comparison of air temperatures. Meanwhile, with increasing air temperature from 22° C, the mean heart rate, RSP, and LF/HF ratio increased. Further, the mean heart rate and RSP under different air temperatures was lower at 55 dBA than at 75 dBA in different workloads. As the values are low, one could interpret a lack of arousal in general at the noise level of 55 dB, but the values are higher at the noise level of 75 dB and also increase with air temperature which could be due to noise impacting arousal. As shown in Table 5, with increase of air temperature from 22° C and noise level, LF/HF ratio variations have a uniform trend. Concerning the combined effects of noise and air temperature on the mean heart rate, LF/HF ratio, and RSP, the effect sizes of increasing air temperatures on the mean heart rates, LF/HF, and RSP at 55 dBA were generally lower than those at 75 dBA (Table 1). In other words, in the presence of high noise level of 75 dBA, the increase in air temperature worsened the mean heart rates, LF/HF, and RSP. Under the air temperature of 18 °C, the results showed that, the mean heart rate, LF/HF, and RSP increased, as compared with the air temperature of 22 °C.

**Table 5**
MD ± SE of changes in heart rate and respiration rate in exposure to different noise levels and air temperatures.

| Noise levels | Variables | Air temperatures | | | | | |
|---|---|---|---|---|---|---|---|
| | | 18 °C | 22 °C | 26 °C | 30 °C | ES | p-value |
| **55 dBA** | HRn = 1 (beat/min) | −1.11 ± 1.02 | −1.00 ± 1.97 | 0.54 ± 1.08 | 1.96 ± 0.77 | 0.064 | 0.104 |
| | HRn = 2 (beat/min) | 2.18 ± 0.92 | 1.39 ± 1.82 | 5.13 ± 1.38 | 5.84 ± 1.03 | 0.140 | 0.064 |
| | HRn = 3 (beat/min) | 2.25 ± 0.87 | 3.51 ± 2.25 | 5.21 ± 1.48 | 7.35 ± 1.28 | 0.058 | 0.116 |
| | LF/HFn = 1 (%) | 2.03 ± 0.85 | 0.45 ± 1.12 | 1.64 ± 0.58 | 1.57 ± 0.52 | 0.018 | 0.567 |
| | LF/HFn = 2 (%) | 2.57 ± 0.89 | 0.54 ± 1.10 | 1.84 ± 0.68 | 2.29 ± 0.58 | 0.035 | 0.297 |
| | LF/HFn = 3 (%) | 5.19 ± 1.05 | 0.94 ± 1.22 | 2.39 ± 0.53 | 6.63 ± 4.55 | 0.033 | 0.298 |
| | RSPn = 1 (bpm) | 2.24 ± 0.57 | 1.01 ± 0.76 | 2.39 ± 0.77 | 3.43 ± 0.48 | 0.060 | 0.061 |
| | RSPn = 2 (bpm) | 2.51 ± 0.72 | 1.12 ± 0.85 | 2.67 ± 0.74 | 4.02 ± 0.63 | 0.071 | 0.067 |
| | RSPn = 3 (bpm) | 3.57 ± 0.67 | 2.05 ± .080 | 3.56 ± 0.75 | 4.87 ± 0.72 | 0.070 | 0.060 |
| 75 dBA | HRn = 1 (beat/min) | 4.92 ± 1.08 | 1.71 ± 0.93 | 6.67 ± 0.87 | 8.79 ± 0.90 | 0.212 | 0.001 |
| | HRn = 2 (beat/min) | 6.81 ± 1.19 | 4.37 ± 1.11 | 9.55 ± 1.03 | 12.31 ± 0.99 | 0.220 | 0.009 |
| | HRn = 3 (beat/min) | 9.74 ± 1.28 | 5.69 ± 1.04 | 12.93 ± 0.77 | 16.05 ± 1.09 | 0.334 | 0.001 |
| | LF/HFn = 1 (%) | 3.34 ± 1.14 | 2.61 ± 0.59 | 3.86 ± 0.51 | 5.21 ± 0.55 | 0.060 | 0.115 |
| | LF/HFn = 2 (%) | 6.00 ± 1.18 | 3.76 ± 0.63 | 5.24 ± 0.62 | 8.34 ± 0.44 | 0.152 | 0.004 |
| | LF/HFn = 3 (%) | 8.33 ± 0.95 | 4.80 ± 0.87 | 7.30 ± 0.64 | 10.58 ± 0.56 | 0.229 | 0.001 |
| | RSPn = 1 (bpm) | 4.20 ± 0.52 | 0.93 ± 0.40 | 3.01 ± 0.48 | 5.29 ± 0.40 | 0.330 | 0.001 |
| | RSPn = 2 (bpm) | 5.74 ± 0.58 | 2.07 ± 0.63 | 3.08 ± 0.68 | 6.52 ± 0.38 | 0.284 | 0.001 |
| | RSPn = 3 (bpm) | 6.83 ± 0.51 | 3.08 ± 0.61 | 5.69 ± 0.59 | 7.38 ± 0.39 | 0.261 | 0.001 |

**Note.** HR: Heart rate changes as compared with the baseline , LF/HF: change in LF to HF ratio as compared with the baseline, RSP: change in respiratory as compared with the baseline, n = 1, n = 2 and n = 3, respectively: low workload, medium workload, and high workload, MD: Mean of difference, SE: Standard error.

**Table 6**
Multiple comparisons of heart rate and respiration rate in three workloads in exposure to different noise levels and air temperatures.

| Noise | Parameters | 18 °C | | | 22 °C | | | 26 °C | | | 30 °C | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 22 °C | 26 °C | 30 °C | 18 °C | 26 °C | 30 °C | 18 °C | 22 °C | 30 °C | 18 °C | 22 °C | 26 °C |
| **55 dBA** | HRn = 1 (beat/min) | NS | **0.011** | **0.010** | NS | NS | NS | **0.011** | NS | NS | **0.010** | NS | NS |
| | HRn = 2 (beat/min) | NS | NS | **0.011** | NS | NS | **0.034** | NS | NS | NS | **0.011** | **0.034** | NS |
| | HRn = 3 (beat/min) | NS | NS | **0.001** | NS | NS | NS | NS | NS | NS | **0.001** | NS | NS |
| | LF/HFn = 1 (%) | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| | LF/HFn = 2 (%) | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| | LF/HFn = 3 (%) | **0.010** | **0.011** | NS | **0.010** | NS | NS | **0.011** | NS | NS | NS | NS | NS |
| | RSPn = 1 (bpm) | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| | RSPn = 2 (bpm) | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS | NS |
| | RSPn = 3 (bpm) | NS | NS | NS | NS | NS | **0.020** | NS | NS | NS | NS | **0.020** | NS |
| **75 dBA** | HRn = 1 (beat/min) | **0.030** | NS | **0.017** | **0.030** | **0.001** | **0.001** | NS | **0.001** | NS | **0.017** | **0.001** | NS |
| | HRn = 2 (beat/min) | NS | NS | **0.001** | NS | **0.003** | **0.001** | NS | **0.003** | NS | **0.002** | **0.001** | NS |
| | HRn = 3 (beat/min) | **0.013** | NS | **0.001** | **0.013** | **0.001** | **0.001** | NS | **0.001** | **0.034** | **0.001** | **0.001** | **0.034** |
| | LF/HFn = 1 (%) | NS | NS | NS | NS | NS | **0.001** | NS | NS | NS | NS | **0.001** | NS |
| | LF/HFn = 2 (%) | NS | NS | NS | NS | NS | **0.001** | NS | NS | **0.001** | NS | **0.001** | **0.001** |
| | LF/HFn = 3 (%) | **0.018** | NS | **0.028** | **0.018** | **0.030** | **0.001** | NS | **0.030** | **0.001** | **0.028** | **0.001** | **0.001** |
| | RSPn = 1 (bpm) | **0.001** | NS | NS | **0.001** | **0.001** | **0.001** | NS | **0.001** | **0.001** | NS | **0.001** | **0.001** |
| | RSPn = 2 (bpm) | **0.001** | NS | NS | **0.001** | **0.007** | **0.001** | NS | **0.007** | **0.034** | NS | **0.001** | **0.037** |
| | RSPn = 3 (bpm) | **0.001** | NS | NS | **0.001** | **0.007** | **0.001** | NS | **0.007** | **0.034** | NS | **0.001** | **0.034** |

Note. The reported values indicating significant difference ($P < 0.05$) and NS indicating no significant.

**Table 7**
Pair wise comparison of the combined effects of noise levels and air temperatures on changes in heart rate and respiration rate.

| Noise levels -air temperatures | HR (beat/min) | | | LF/HF (%) | | | RSP (bpm) | | |
|---|---|---|---|---|---|---|---|---|---|
| | MD | ES | p | MD | ES | p | MD | ES | p |
| 75dBA-22 °C vs. 55dBA-22°Cn = 1 | 2.72 ± 1.78 | 0.064 | 0.137 | 2.16 ± 1.26 | 0.079 | 0.097 | −0.76 ± 0.90 | 0.001 | 0.934 |
| 75dBA-30 °C vs. 55dBA-30°Cn = 1 | 6.48 ± 1.10 | 0.503 | 0.001 | 3.63 ± 0.42 | 0.679 | 0.001 | 1.85 ± 0.48 | 0.302 | 0.001 |
| 75dBA-22 °C vs. 55dBA-22°Cn = 2 | 2.98 ± 1.84 | 0.071 | 0.115 | 3.22 ± 1.23 | 0.168 | 0.013 | 0.94 ± 0.98 | 0.026 | 0.344 |
| 75dBA-30 °C vs. 55dBA-30°Cn = 2 | 6.47 ± 1.27 | 0.433 | 0.001 | 6. 04 ± 0.53 | 0.788 | 0.001 | 2.49 ± 0.64 | 0.307 | 0.001 |
| 75dBA-22 °C vs. 55dBA-22°Cn = 3 | 2.17 ± 2.20 | 0.028 | 0.331 | 3.85 ± 1.51 | 0.160 | 0.015 | 1.02 ± 0.89 | 0.037 | 0.256 |
| 75dBA-30 °C vs. 55dBA-30°Cn = 3 | 8.69 ± 1.32 | 0.559 | 0.001 | 3.93 ± 4.40 | 0.023 | 0.371 | 2.50 ± 0.71 | 0.266 | 0.001 |

**Note.** MD: Mean of difference (pre- and post-), SE: Standard error, ES: Effect size.

The effect sizes presented in Tables 7 and 8 demonstrate the significant effect of high air temperature (55 dBA, 30 °C) on heart rate and RSP, as compared with the high noise level (75 dBA, 22 °C) in the three workloads. Table 4 also presents the combined effect of noise and temperature (75 dBA at 30 °C) on heart rate, RSP, and LF/HF which was more significant than the effect of each of them individually. Pair wise comparisons of the combined effects of noise levels and air temperatures on the changes in heart rate and respiration rate are presented in Tables 7 and 8 As shown in Table 5, the two indices of HR and RSP had a uniform trend under different noise levels and air temperatures.

Table 9 presents the values of theta, alpha, and beta amplitudes and eye electrical activity before and after the exposure to different scenarios. It is observed that significant differences exsist between the mean values of theta, alpha, and beta amplitudes and eye electrical activity at different air temperatures with noise levels of 55 dBA and 75 dBA ($P < 0.05$). Table 10 also shows multiple comparisons of the electroencephalograph and electrooculograph amplitude of different bands under coexposure of noise and air temperature. Generally, with increasing air temperatures and noise levels, there was a change in mean theta amplitude and eye electrical activity and a decrease in mean alpha and beta amplitudes. Based on the observed uniform trends in relation to increase in air temperatures and noise levels with brain activities, it can be stated that there is a significant correlation between the brain activity data as main indicator of arousal level and working memory. However, the correlation between these indices can be investigated in complementary statistical analysis.

Concerning the combined effects of noise and air temperature on mean theta, alpha, and beta amplitudes, the effect size of increasing air temperatures on mean theta, alpha, and beta amplitudes at 75 dBA was generally higher than that at 55 dBA. In other words, in the presence of high noise level of 75 dBA, the increase in air temperature worsened the mean theta, alpha, and beta amplitudes. However, the mentioned trend was not observed in eye electrical activity.

The effect of noise at 75 dBA and 22 °C on brain electrical activity in theta and alpha amplitudes was less than that at 55 dBA and 30 °C. However, in other conditions, these relationships were not the same for

**Table 8**
Pair wise comparison of the combined effects of air temperatures and different noise levels on changes in heart rate and respiration rate.

| Air temperature- noise levels | HR (beat/min) | | | LF/HF (%) | | | RSP(pbm) | | |
|---|---|---|---|---|---|---|---|---|---|
| | MD | ES | p | MD | ES | p | MD | ES | p |
| 30°C-55dBA vs.22°C-55dBAn = 1 | 4.62 ± 1.61 | 0.171 | 0.012 | 2.33 ± 1.21 | 0.098 | 0.63 | 2.43 ± 0.90 | 0.176 | 0.011 |
| 30°C-75dBA vs.22°C-75dBAn = 1 | 9.08 ± 1.28 | 0.596 | 0.001 | 4.51 ± 1.23 | 0.282 | 0.001 | 4.47 ± 0.49 | 0.706 | 0.001 |
| 30°C-55dBA vs.22°C-55dBAn = 2 | 6.06 ± 1.83 | 0.244 | 0.001 | 2.92 ± 1.14 | 0.162 | 0.015 | 2.90 ± 1.01 | 0.195 | 0.007 |
| 30°C-75dBA vs.22°C-75dBAn = 2 | 8.56 ± 1.31 | 0.555 | 0.001 | 5.83 ± 0.99 | 0.502 | 0.001 | 5.24 ± 0.50 | 0.760 | 0.001 |
| 30°C-55dBA vs.22°C-55dBAn = 3 | 5.25 ± 1.25 | 0.341 | 0.001 | 7.30 ± 0.45 | 0.455 | 0.118 | 2.82 ± 0.90 | 0.224 | 0.004 |
| 30°C-75dBA vs.22°C-75dBAn = 3 | 11.63 ± 1.27 | 0.711 | 0.001 | 9.37 ± 1.04 | 0.705 | 0.001 | 5.78 ± 0.61 | 0.720 | 0.001 |

**Note.** MD: Mean of difference (pre- and post-), SE: Standard error, ES: Effect size.

**Table 9**
Statistical analysis of global electroencephalograph and electrooculograph amplitude of different bands in exposure to different noise levels and air temperatures (MD± SE).

| Noise levels | Variables | Air temperatures | | | | | |
|---|---|---|---|---|---|---|---|
| | | 18 °C | 22 °C | 26 °C | 30 °C | ES | p-value |
| **55 dBA** | Theta (μv) | 1.24 ± 0.27 | 1.74 ± 0.26 | 2.91 ± 0.31 | 3.87 ± 0.42 | 0.260 | 0.001 |
| | alpha (μv) | −0.65 ± 0.23 | 0.82 ± 0.21 | −2.18 ± 0.32 | −2.09 ± 0.38 | 0.486 | 0.001 |
| | Beta(μv) | 1.48 ± 0.16 | 1.19 ± 0.22 | −2.96 ± 0.37 | −3.17 ± 0.15 | 0.773 | 0.001 |
| | Eye activity (μv) | 2.44 ± 0.85 | 3.04 ± 1.20 | 6.42 ± 1.07 | 8.13 ± 1.59 | 0.134 | 0.002 |
| **75 dBA** | Theta (μv) | 1.77 ± 0.23 | 1.59 ± 0.15 | 3.35 ± 0.23 | 4.89 ± 0.38 | 0.512 | 0.001 |
| | alpha (μv) | −1.42 ± 0.16 | −2.04 ± 0.27 | −3.88 ± 0.19 | −4.72 ± 0.22 | 0.592 | 0.001 |
| | Beta(μv) | 0.38 ± 0.30 | −3.69 ± 0.18 | −4.87 ± 0.25 | −6.20 ± 0.17 | 0.814 | 0.001 |
| | Eye activity (μv) | 5.46 ± 0.97 | 8.36 ± 1.32 | 10.54 ± 0.76 | 12.10 ± 0.81 | 0.202 | 0.001 |

**Note.** MD: Mean of difference (pre- and post-) , SE: Standard error, ES: Effect size.

**Table 10**
Multiple comparisons of the electroencephalograph and electrooculograph amplitude of different bands in exposure to different noise levels and air temperatures.

| Noise | Parameters | 18 °C | | | 22 °C | | | 26 °C | | | 30 °C | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 22 °C | 26 °C | 30 °C | 18 °C | 26 °C | 30 °C | 18 °C | 22 °C | 30 °C | 18 °C | 22 °C | 26 °C |
| **55 dBA** | Theta (μv) | NS | **0.001** | **0.001** | NS | **0.013** | **0.001** | **0.001** | **0.013** | NS | **0.001** | **0.001** | NS |
| | alpha (μv) | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | NS | **0.001** | **0.001** | NS |
| | Beta(μv) | NS | **0.001** | **0.001** | NS | **0.001** | **0.001** | **0.001** | **0.001** | NS | **0.001** | **0.001** | NS |
| | Eye activity (μv) | NS | **0.008** | **0.002** | NS | **0.035** | **0.013** | **0.008** | **0.035** | NS | **0.004** | **0.013** | NS |
| **75 dBA** | Theta (μv) | NS | **0.001** | **0.001** | NS | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** |
| | alpha (μv) | NS | **0.001** | **0.001** | NS | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** |
| | Beta(μv) | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** | **0.001** |
| | Eye activity(μ) | **0.027** | **0.001** | **0.001** | **0.027** | NS | **0.020** | **0.001** | NS | NS | **0.001** | **0.020** | NS |

**Note.** The reported values indicating significant difference ($P < 0.05$) and NS indicating no significant.

beta amplitude and eye electrical activity. Beta activity generally decreased with increased air temperature and is related to alertness, attentiveness, and focused mental activity. The results also indicate that the combined effects of noise and air temperature at 75 dBA at 30 °C on alpha, beta, and theta amplitudes and eye electrical activity were more significant than the influence of each of them alone.

## 4. Discussion

In order to provide an acceptable indoor workroom, first each of the environmental factors should be at a satisfactory level, while the adverse effect of interactions between these factors is restricted (ASHRAE Guideline 10–2016).The goal of this study was to add new data about the interaction effect of noise and air temperature on working memory and neurophysiological responses under different mental workloads.

The current results generally confirmed that the participants experienced fairly optimum value of arousal at 22 °C, hence this air temperature resulted in higher level of accuracy, as compared with other air temperatures. This finding is in agreement with the result reported by Lan et al. (2009) and Cui et al. (2013). Compared with the other thermal conditions as 18 °C, 26 °C, and 30 °C, the optimum values for working memory and physiological responses were also observed while the participants were exposed to 22 °C (see Tables 1 and 5). It may be argued that the relatively neutral condition decreases the arousal up to an optimal range. It was observed that when noise and air temperature reached an optimum level, the effects of noise and air temperature on working memory and psychological indices decreased. Then, with a further increase, going beyond the neutral state, its effect increased. Hence, the temperatures of 18 °C and 30 °C had a more significant effect on working memory and neurophysiological indices.

To achieve an optimum level of task performance, it is necessary to have a certain level of stimulation or arousal. Yerkes–Dodson law dictates that work performance improves with arousal/alertness up to an optimal point beyond which the work performance decreases (Yerkes and Dodson, 1908). Our findings confirmed that exposure to extreme thermal conditions and high noise level may lead to arousal increase to a level where cognitive performance is degraded. This finding is consistent with the results of Seppanen et al. (2003), who proposed a relationship between performance and air temperature illustrating a decline in performance by 2% per 1 °C increase in the air temperature within the range of 25–32°*C*. On the other hand, it had no impact on performance within the air temperature range of 21–25.°*C*.

It is observed that air temperature elevation beyond 22 °C, the mean level of changes in the heart rate and respiration rate as well as LF/HF ratio also increased. Previous researches showed the rise in these indices can cause an increase in participants' mental workload and stress (Widyanti et al., 2013; Hwang et al., 2008). Smit et al. (2015) and Munzel et al. (2014) also reported changes in response to autonomic indices such as blood pressure, heart rate, and respiration rate during exposure to extreme environmental conditions. The EEG analysis showed that with the increase in air temperatures and noise levels, the theta indicators increased while the alpha and beta indices decreased, which can lead to occupational burnout in workers in the long term. The same findings were also reported by Cho at al. (2011), Azemi et al. (2009) and Smith et al. (2002). The results of this study further verified that the changes in theta, alpha, and beta amplitudes and eye electrical activity under the noise level of 55 dBA were less considerable than the variations under 75 dBA. The beta amplitude at 18 °C and 75 dBA was higher than that at 22 °C and 75 dBA, indicating that people at this relatively low air temperature experienced more mental stress. Some studies have also reported the effect of air temperature (Yao et al., 2008; Tham and Willem, 2010) and noise (Cho et al., 2011; Zhou et al., 2012) on brain electrical activities. Choi et al. (2015) showed that based on brain electrical activities after exposure to road traffic noise and high air temperature conditions, high stress was imposed on the studied subjects. Note that the increase in alpha amplitude improves cognitive performance, while its reduction indicates burden of mental workload and high stress (Doppelmayr et al., 2007).

The findings strongly suggest that the effect of noise on working memory is stronger than that of air temperature in the current experiment

configurations with the included air temperatures (18–30 °C) and noise levels (55 dBA and 75 dBA), which is congruent with the results obtained by Pellerin and Candas (2004). Some studies have shown that memory is very susceptible to interference by noise, especially when cognitive tasks are related to the maintenance of order short-term working memory (Banbury et al., 2001; Banis and Lorist, 2012). In the presence of high noise level, the increase in air temperature did not worsen the response accuracy. It can be stated that high noise neutralized the effect of increasing air temperatures on response accuracy. Furthermore, similar findings were observed about determining the effect size of increasing noise levels on the response accuracy under high air temperatures. Nagano and Horikoshi (2005) reported that the noisy sensation and quiet sensation were significantly affected by air temperature. However, in the presence of high noise level, the increase in air temperature resulted in aggravated neurophysiological responses. Witterseh et al. (2004) also reported that combined effects were found between thermal conditions and noise. Moreover, the effects of air temperature on heart rate and respiratory rate indices were more significant than those of noise in the three workload conditions. Note that, considering the effect size of workload levels on working memory and neurophysiological aspects, work complexity does not interact with noise and air temperature. In addition, the effect of increase in air temperature on the mean changes in theta, alpha and beta amplitudes was more significant than the effect of exposure to a noise level of up to 75 dB.

As with any other research, this study suffered from some limitations. The short-term exposures to different scenarios of environmental conditions may have resulted in restricted human responses. Further, field studies are required to show cognitive performance in occupations that require long-term exposures in extreme environmental conditions. It is also suggested future researches focus on the effect of gender as an important variable. It is proposed that some aspects of the interaction effects of these external stressors be investigated by more advanced methods (e.g. neuroimaging). The current study focused on the balance between air temperature and noise in terms of their effects on working memory and psychophysiological indices. It is now one of the prerequisites for the design of indoor environments, such as offices, control rooms, and other similar environments. The results of this study provide enough evidence to modify the managers' outlook, policies, and practices in designing workrooms and setting environmental comfort standards for office buildings.

## 5. Conclusion

In order to design an acceptable indoor environment, not only should each environmental parameter, including noise, lighting, and air temperature be in acceptable level, but also the combined impacts of these parameters must be considered. This study evaluated the mechanism through which mental stress was affected by to two main air environmental factors based on a neurophysiological approach. It was found that this approach could be used as a basis for future multidimensional studies. The following conclusions can be drawn from the results of the current study which are limited to the experiment configurations of air temperatures (18–30 °C) and noise levels (55 dBA and 75 dBA).

- As air temperatures increased beyond the neutral state and with noise elevation, working memory and the neurophysiological responses were disturbed. The increase in noise had a significant effect on response accuracy, as compared with air temperature.
- In the presence of high noise level, the increase in air temperature did not worsen the accuracy. Similar results were also observed when increasing noise levels in high air temperatures. However, in the presence of high noise levels, the increase in air temperature aggravated the mean value of neurophysiological responses.
- High air temperature had a significant effect on neurophysiological response, as compared with the high noise in the three workloads. However, the same finding was not observed for the beta band and eye electrical activity.
- As predicted, the combined effects of noise and air temperature on working memory and psychophysiological indices were more significant than the influence of each of them alone. Based on the effect size of workloads on working memory and neurophysiological aspects, work complexity did not interact with noise and air temperature.
- The more detailed analysis indicated that the participants felt more comfortable at 22 °C and 55dBA, as compared with the other conditions.

Generally, air temperature and noise had an interaction effect on working memory and neurophysiological responses. Therefore, authorities must prioritize the modification of indoor environment in order to manage these factors. Many future research should be carried out to investigate the neurobehavioral trends of coexposure to physical agents in indoor environment.

## Declaration of competing interest

There is no conflict of interest.

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